

Sign-Based Optimizers Under Heavy-Tailed Noise: On the Effectiveness of Lion and Muon

Dingzhi Yu^{1,2} Hongyi Tao^{1,2} Yuanyu Wan³ Luo Luo⁴ Lijun Zhang^{1,2}

¹State Key Laboratory of Novel Software Technology, Nanjing University ²School of Artificial Intelligence, Nanjing University ³School of Software Technology, Zhejiang University ⁴Fudan University



1 Why do sign-based optimizers win?

- AdamW is the de facto LLM optimizer, yet sign-based Lion and Muon consistently beat it in LLM pre- and post-training.
- We still lack a theory of *why* they outperform AdamW.
- Their edge is largest in **LLM training**, where gradient noise is heavy-tailed, and shrinks in **vision** tasks, whose gradient noise is typically more concentrated.
- Heavy tails break the usual tools: SGD can diverge in $\mathbb{E}\|\nabla f(\mathbf{x}_t)\|_2^2$ when $p < 2$ [Zhang+20a, Fatkhullin+25]; Adam/AdaGrad have no rigorous heavy-tail theory and could even converge worse [Chezhegov+25].

2 LLM gradient noise is heavy-tailed

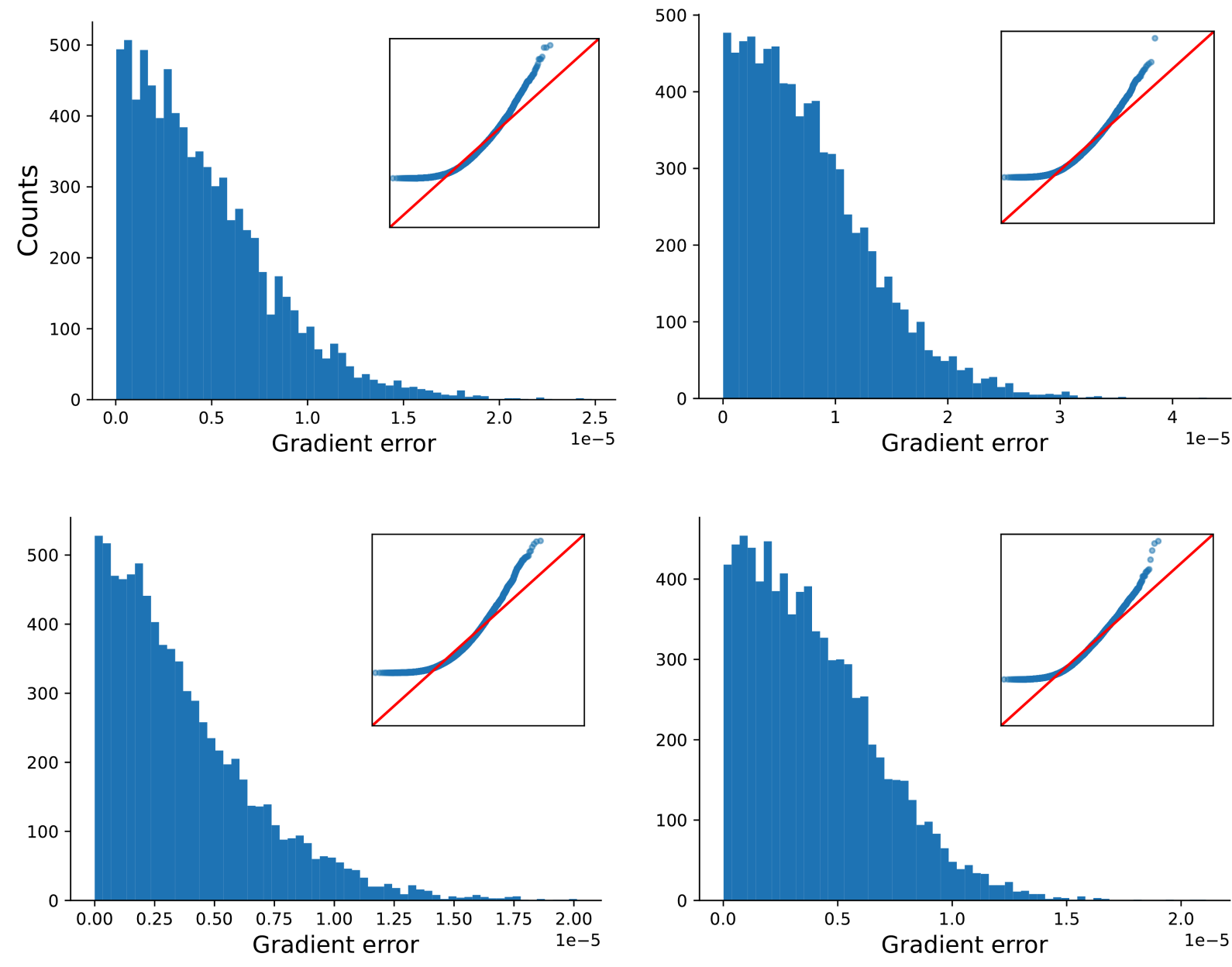


Fig. 1. GPT-2 pretraining on C4, at initialization. Histograms of gradient errors at randomly sampled coordinates; the inset Q-Q plots against a Gaussian reference show pronounced tail deviations.

Heavy-tailed regime: the noise has only a finite p -th moment, $p \in (1, 2]$; $p = 2$ is the finite-variance case.

- Classic model** ($\sigma_1 = \mathbf{0}$): $\mathbb{E}|g_i - \nabla_i f| \leq \sigma_{0,i}^p$. It might fail even on simple linear regression tasks.
- Ours:** $\mathbb{E}|g_i - \nabla_i f|^p \leq \sigma_{0,i}^p + \sigma_{1,i}^p |\nabla_i f|^p$, so the noise level is allowed to grow with the gradient.

3 Question & our answer

Can we obtain theoretical guarantees of sign-based optimizers to explain their practical effectiveness and justify their advantage over AdamW under heavy-tailed gradient noise?

Yes. New convergence theory for vector (SignSGD, Lion) and matrix (Muon, Muonlight) sign descent:

- Generalized heavy-tailed noise** for vectors and matrices: the p -th moment may grow with the gradient; strictly weaker than existing assumptions and supported by GPT-2 measurements.
- Sharp rates** $O(T^{-\frac{p-1}{3p-2}})$ for SignSGD, Lion, Muon and Muonlight under generalized smoothness, matching or surpassing the previous state of the art.
- GPT-2 pretraining** validates the noise model; full-model curvature and noise measurements, with matrix effective ranks, reveal geometry that favors our bounds over prior Euclidean and Frobenius analyses.

Our results suggest the sign operation (element-wise or via orthogonalization) acts as a **natural robustifier** against heavy-tailed noise, which makes sign-based optimizers like Lion and Muon a natural fit for training LLMs. Against normalized SGD the advantage is provable (see 6).

One sharp rate for all four sign optimizers

$$O\left(T^{-\frac{p-1}{3p-2}}\right)$$

SignSGD · Lion · Muon · Muonlight under generalized heavy-tailed noise ($p \in (1, 2]$) and generalized smoothness (conditions in the theorem cards). Complexity $O(\epsilon^{-\frac{3p-2}{p-1}})$, recovering $O(\epsilon^{-4})$ at $p = 2$; matches or surpasses the best-known bounds.

4 VECTOR SIGN OPTIMIZERS SignSGD & Lion

Alg. 1 SignSGD [Bernstein+18]

```
1 Input:  $T, \mathbf{x}_1, \eta, \mathbf{m}_0 := \mathbf{g}_1$ 
2 for  $t = 1$  to  $T$  do
3    $\mathbf{g}_t = \frac{1}{B} \sum_{b=1}^B \mathbf{g}_t^b$ 
4    $\mathbf{m}_t = \beta \mathbf{m}_{t-1} + (1 - \beta) \mathbf{g}_t$ 
5    $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \text{sign}(\mathbf{m}_t)$ 
```

Alg. 2 Lion [Chen+23]

```
1 Input:  $T, \mathbf{x}_1, \eta, \lambda; \mathbf{m}_0 := \mathbf{g}_1$ 
2 for  $t = 1$  to  $T$  do
3    $\mathbf{g}_t = \frac{1}{B} \sum_{b=1}^B \mathbf{g}_t^b$ 
4    $\mathbf{v}_t = \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1) \mathbf{g}_t$ 
5    $\mathbf{m}_t = \beta_2 \mathbf{m}_{t-1} + (1 - \beta_2) \mathbf{g}_t$ 
6    $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \text{sign}(\mathbf{v}_t) - \eta \lambda \mathbf{x}_t$ 
```

Lion, found by program search, decouples the update momentum $\mathbf{v}_t$ from the stored $\mathbf{m}_t$ and adds decoupled weight decay; it reduces to SignSGD when $\beta_1 = \beta_2$.

A Assumptions 1a–4a (vector $\mathbf{x} \in \mathbb{R}^d$)

1a Lower bounded: f possibly non-convex, a.e. twice-differentiable, $f^* := \inf_{\mathbf{x}} f(\mathbf{x}) > -\infty$.

2a ($\mathbf{l}_0, \mathbf{l}_1$)-vector smoothness: $\exists \mathbf{l}_0, \mathbf{l}_1 \in \mathbb{R}_+^d$ s.t. for all $\|\mathbf{x}' - \mathbf{x}\|_\infty \leq 1/\|\mathbf{l}_1\|_\infty$,

$$|f(\mathbf{x}') - f(\mathbf{x}) - \langle \nabla f(\mathbf{x}), \mathbf{x}' - \mathbf{x} \rangle| \leq \frac{1}{2} \|\mathbf{x}' - \mathbf{x}\|_{\mathbf{l}_0 + \mathbf{l}_1 \odot |\nabla f(\mathbf{x})|}^2,$$

with the weighted norm $\|\mathbf{x}\|_1^2 := \mathbf{x}^\top \text{diag}(\mathbf{l}) \mathbf{x}$.

3a Unbiasedness: B mutually independent gradients per step with $\mathbb{E}[\mathbf{g}_t^b | \mathcal{F}_{t-1}] = \nabla f(\mathbf{x}_t)$.

4a (σ_0, σ_1)-vector heavy-tailed noise: $\exists p \in (1, 2], \sigma_0, \sigma_1 \in \mathbb{R}_+^d$ s.t. $\forall i$,

$$\mathbb{E}[|g_{t,i}^b - \nabla_i f(\mathbf{x}_t)|^p | \mathcal{F}_{t-1}] \leq \sigma_{0,i}^p + \sigma_{1,i}^p |\nabla_i f(\mathbf{x}_t)|^p.$$

$\mathbf{l}_1 = \mathbf{0}$: coordinate-wise smoothness used for SignSGD [Bernstein+18]; $\mathbf{l}_1 \succ \mathbf{0}$ extends it to generalized smoothness [Zhang+20b], which aligns with practice. $\sigma_1 = \mathbf{0}$ recovers the classic coordinate-wise heavy-tailed noise model [Kornilov+25].

Thm Theorems 1 & 2: SignSGD and Lion

Under A1a–4a, with $\Delta_f := f(\mathbf{x}_1) - f^*, \beta, \eta$ as in the paper and batch $B = \max\{1, \lceil (c\|\sigma_1\|_\infty)^{\frac{p}{p-1}} \rceil\}$ ($c = 64\sqrt{2}$ for SignSGD, $144\sqrt{2}$ for Lion):

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \|\nabla f(\mathbf{x}_t)\|_1 \leq O\left((\Delta_f \|\mathbf{l}_0\|_1)^{\frac{p-1}{3p-2}} \|\sigma_0\|_1^{\frac{p}{3p-2}} (BT)^{-\frac{p-1}{3p-2}}\right)$$

If $\|\mathbf{x}_1\|_\infty \leq \frac{1}{3\lambda}$, Lion attains this rate for any $\beta_1 \in [1 - (1 - \beta_2)^{\frac{p-1}{p}}, 1]$ and $\lambda \in [0, (1 - 2^{-1/T})/\eta]$, with $\|\mathbf{x}_t\|_\infty \leq \frac{2}{3\lambda}$ for all t (uniform iterate bound).

Complexity $O(\epsilon^{-\frac{3p-2}{p-1}})$ matches the ℓ_2 lower bound, with no explicit dependence on d and any $\mathbf{l}_1, \sigma_1 \geq \mathbf{0}$. To our knowledge, **the first heavy-tail guarantee for (unclipped) Lion**.

vs Prior vector analyses

- Kornilov+25 ($\sigma_1 = \mathbf{0}$): extra $d^{\frac{3p-2}{2(p-1)}} / (\phi_1 \phi_\sigma^{\frac{p}{p-1}})$; up to $d^{\frac{3p-2}{2(p-1)}}$ is saved for sparse $\mathbf{l}_0, \sigma_0$, and even their best case still pays an extra factor $d^{\frac{2-p}{2(p-1)}}$.
- Jiang+26 (Lion, $p = 2$): same isotropic worst case, but ours scales with $\|\sigma_0\|_1 \leq \sqrt{d} \|\sigma_0\|_2$ and $\|\mathbf{l}_0\|_1 \leq d \|\mathbf{l}_0\|_\infty$; milder $\beta_1, \lambda, \mathbf{x}_1$ conditions.
- Tightness ($p = 2, \mathbf{l}_1 = \sigma_1 = \mathbf{0}$): our stochastic term matches the ℓ_1 lower bound of Tao+26 for SignSGD without momentum (finite-variance subclass).

6 Provable gain over normalized gradient descent (to find an ϵ -stationary point)

Setting	Algorithm	Criterion	Complexity	Improvement
Vector	NSGD [Liu & Zhou 25]	$\mathbb{E}\ \nabla\ _2 \leq \epsilon$	$O\left(\frac{\Delta_f \ \mathbf{l}_0\ _\infty \ \sigma_0\ _2^{\frac{p}{p-1}}}{\epsilon^{\frac{3p-2}{p-1}}}\right)$	= lower bound
	SignSGD & Lion (ours)	$\mathbb{E}\ \nabla\ _1 \leq \epsilon$	$O\left(\frac{\Delta_f \ \mathbf{l}_0\ _1 \ \sigma_0\ _1^{\frac{p}{p-1}}}{\epsilon^{\frac{3p-2}{p-1}}}\right)$	up to d
Matrix	MNSGD [Liu & Zhou 25]	$\mathbb{E}\ \nabla\ _F \leq \epsilon$	$O\left(\frac{\Delta_f \ \mathbf{L}_0\ _{\text{op}} \ \mathbf{V}_0\ _F^{\frac{p}{p-1}}}{\epsilon^{\frac{3p-2}{p-1}}}\right)$	= lower bound
	Muon & Muonlight (ours)	$\mathbb{E}\ \nabla\ _* \leq \epsilon$	$O\left(\frac{\Delta_f \ \mathbf{L}_0\ _* \ \mathbf{V}_0\ _*^{\frac{p}{p-1}}}{\epsilon^{\frac{3p-2}{p-1}}}\right)$	up to $\min\{m, n\}$

With $\mathbf{l}_1 = \mathbf{L}_1 = \sigma_1 = \mathbf{V}_1 = \mathbf{0}$; the NSGD/MNSGD rates are tight.

5 MATRIX SIGN OPTIMIZERS Muon & Muonlight

Alg. 3 Muon [Jordan+24]

```
1 Input:  $T, \mathbf{X}_1, \eta; \mathbf{B}_0 := \mathbf{0}$ 
2 for  $t = 1$  to  $T$  do
3    $\mathbf{G}_t = \frac{1}{B} \sum_{b=1}^B \mathbf{G}_t^b$ 
4    $\mathbf{B}_t = \beta \mathbf{B}_{t-1} + \mathbf{G}_t$ 
5    $\mathbf{O}_t = \text{NewtonSchulz}(\mathbf{B}_t)$ 
6    $\mathbf{X}_{t+1} = \mathbf{X}_t - \eta \mathbf{O}_t$ 
```

Alg. 4 Muonlight [Liu+25]

```
1 Input:  $T, \mathbf{X}_1, \eta, \lambda; \mathbf{B}_0 := \mathbf{0}$ 
2 for  $t = 1$  to  $T$  do
3    $\mathbf{G}_t = \frac{1}{B} \sum_{b=1}^B \mathbf{G}_t^b$ 
4    $\mathbf{B}_t = \beta_2 \mathbf{B}_{t-1} + \mathbf{G}_t$ 
5    $\tilde{\mathbf{B}}_t = \beta_1 \mathbf{B}_t + \mathbf{G}_t$ 
6    $\mathbf{O}_t = \text{NewtonSchulz}(\tilde{\mathbf{B}}_t)$ 
7    $\mathbf{X}_{t+1} = \mathbf{X}_t - \eta \mathbf{O}_t - \eta \lambda \mathbf{X}_t$ 
```

Muonlight adds Nesterov momentum and decoupled weight decay; $\text{msign}(\mathbf{X}) = \mathbf{U}\mathbf{V}^\top$.

A Assumptions 1b–4b (matrix $\mathbf{X} \in \mathbb{R}^{m \times n}$)

1b Lower bounded: f possibly non-convex, $f^* := \inf_{\mathbf{X}} f(\mathbf{X}) > -\infty$.

2b ($\mathbf{L}_0, \mathbf{L}_1$)-matrix smoothness: $\exists \mathbf{L}_0, \mathbf{L}_1 \in \mathbb{R}^{m \times n}$, $\mathbf{L}_0$ full row rank, s.t. for all $\|\mathbf{X}' - \mathbf{X}\|_{\text{op}} \leq 1/\|\mathbf{L}_1\|_{\text{op}}$,

$$\|\nabla f(\mathbf{X}') - \nabla f(\mathbf{X})\|_{(\mathbf{L}(\mathbf{X}))^{-1}} \leq \|\mathbf{X}' - \mathbf{X}\|_{\mathbf{L}(\mathbf{X})}, \quad \mathbf{L}(\mathbf{X}) = |\mathbf{L}_0|_{\text{m}} + |\nabla f(\mathbf{X}) \mathbf{L}_1^\top|_{\text{m}}.$$

3b Unbiasedness: B mutually independent gradients per step with $\mathbb{E}[\mathbf{G}_t^b | \mathcal{F}_{t-1}] = \nabla f(\mathbf{X}_t)$.

4b ($\mathbf{V}_0, \mathbf{V}_1$)-matrix heavy-tailed noise: $\exists p \in (1, 2], \mathbf{V}_0, \mathbf{V}_1 \in \mathbb{R}^{m \times n}$, $\mathbf{V}_0$ full row rank, s.t.

$$\mathbb{E}[\|\mathbf{V}_0\|_*^{\frac{p}{p-1}} \|\mathbf{G}_t^b - \nabla f(\mathbf{X}_t)\|_{\mathbf{V}_0}^p | \mathcal{F}_{t-1}] \leq \|\mathbf{V}_0\|_*^p + |\langle \mathbf{V}_1, \nabla f(\mathbf{X}_t) \rangle|^p.$$

$|\mathbf{X}|_{\text{m}} := (\mathbf{X}\mathbf{X}^\top)^{1/2}$, $|\mathbf{X}|_{\text{L}}^2 := \text{tr}(\mathbf{X}^\top \mathbf{L} \mathbf{X})$. Like coordinate-wise bounds for vectors, the $|\mathbf{V}_0|_{\text{m}}^2$ -weighted norm captures the structure of the noise matrix; $\mathbf{V}_1 = \mathbf{0}, p = 2$ reduces it to the adaptive-variance condition $\mathbb{E}\|\mathbf{G}_t^b - \nabla f(\mathbf{X}_t)\|_{\mathbf{V}_0}^2 \leq \|\mathbf{V}_0\|_*$ [Kovalev 25, Xie+ 26].

Thm Theorems 3 & 4: Muon and Muonlight

Under A1b–4b and exact Newton–Schulz, with $\Delta_f := f(\mathbf{X}_1) - f^*, \beta, \eta$ as in the paper and batch $B = \max\{1, \lceil (c\|\mathbf{V}_1\|_{\text{op}})^{\frac{p}{p-1}} \rceil\}$ ($c = 64\sqrt{2}$ for Muon, 5958 for Muonlight):

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E} \|\nabla f(\mathbf{X}_t)\|_* \leq O\left((\Delta_f \|\mathbf{L}_0\|_*)^{\frac{p-1}{3p-2}} \|\mathbf{V}_0\|_*^{\frac{p}{3p-2}} (BT)^{-\frac{p-1}{3p-2}}\right)$$

If $\|\mathbf{X}_1\|_{\text{op}} \leq \frac{1}{3\lambda}$, Muonlight attains this rate and $\|\mathbf{X}_t\|_{\text{op}} \leq \frac{2}{3\lambda} \forall t$, for any $\lambda \in [0, (1 - 2^{-1/T})/\eta]$ and $\beta_1 \in [\max\{0, \beta_2 - \frac{2}{20}\}, \min\{1, \beta_2 + \frac{3}{20}\}]$.

To our knowledge, **the first rigorous analysis of matrix optimization under heavy-tailed noise**; for Muonlight, also the first for $p \in (1, 2)$.

vs Prior matrix analyses

- Shen+26 (Frobenius, $p = 2$): same isotropic worst case $r^{3/4}$, $r = \min\{m, n\}$; ours scales with $\|\mathbf{V}_0\|_* \leq \sqrt{r} \|\mathbf{V}_0\|_F$, $\|\mathbf{L}_0\|_* \leq r \|\mathbf{L}_0\|_{\text{op}}$, sharper at low effective rank.
- Sfyraki & Wang 26: clipped Lion/Muon only, $O(T^{-\frac{p-1}{3p-2}} \log T)$, Euclidean norms, bounded gradients, $\lambda > 0$. We cover the standard algorithms without clipping.
- Tightness ($p = 2, \mathbf{L}_1 = \mathbf{V}_1 = \mathbf{0}$): our stochastic term matches the nuclear-norm lower bound of Tao+26 for Muon without momentum (finite-variance subclass).

Where the gain comes from: density

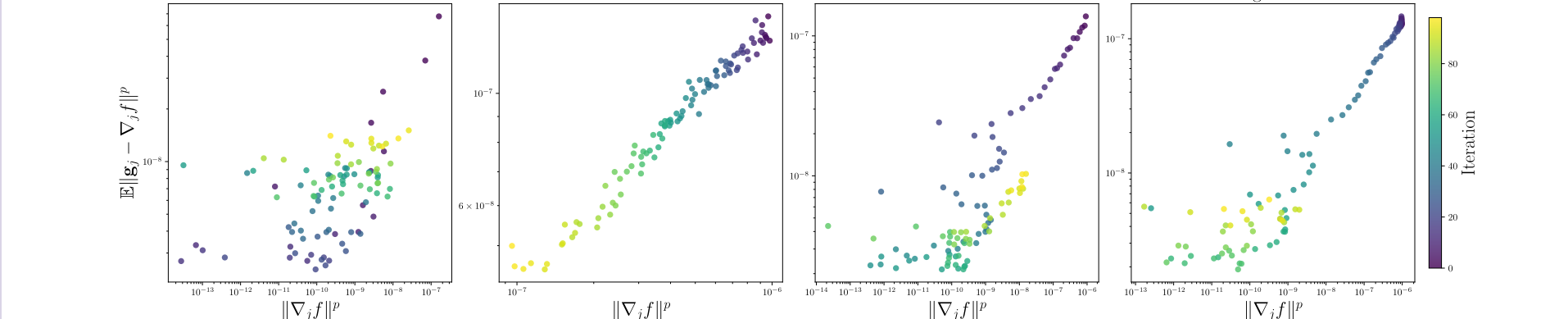
Density (larger = denser): $\phi_q(\mathbf{v}) := \frac{\|\mathbf{v}\|_1}{\|\mathbf{v}\|_q} \in [1, d^{1-\frac{1}{q}}]$; matrices use ψ_q on singular values. In ℓ_2 stationarity, sign vs NSGD complexity is governed by

$$R = R_1 R_2^{\frac{p}{2(p-1)}}, \quad R_1 = \frac{\phi_\infty(\mathbf{l}_0)}{\phi_2^2(\nabla_T)}, \quad R_2 = \frac{\phi_2^2(\sigma_0)}{\phi_2^2(\nabla_T)}$$

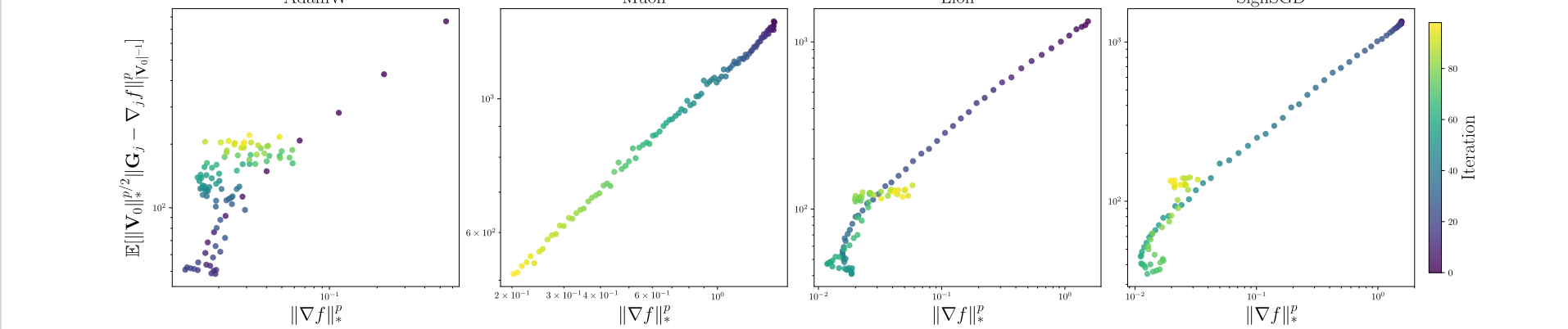
- R_1 : curvature vs gradient density; R_2 : noise vs gradient density (a mild constant); both use the trajectory density $\phi_2(\nabla_T) := \min_t \phi_2(\nabla f(\mathbf{x}_t))$.
- $R < 1$: **our upper bound beats NSGD's lower bound**.
- Dense gradients ($\phi_2(\nabla_T) \approx \sqrt{d}$), axis-aligned curvature ($\phi_\infty(\mathbf{l}_0) \approx 1$): $R_1 \approx 1/d$, a d -fold gain. Muon: low-rank Hessian, high-rank gradients give $R_1 \approx 1/\min\{m, n\}$.

7 The generalized noise model fits

GPT-2 (124M), C4, frozen checkpoints, fresh mini-batches: noise moments grow roughly linearly with the gradient quantity, as $\sigma_1 / \mathbf{V}_1$ allow.



Vector (A4a): $\mathbb{E}|g_j - \nabla_j f|^p$ vs $|\nabla_j f|^p$ near initialization, along AdamW / Muon / Lion / SignSGD runs (colour: training iteration).

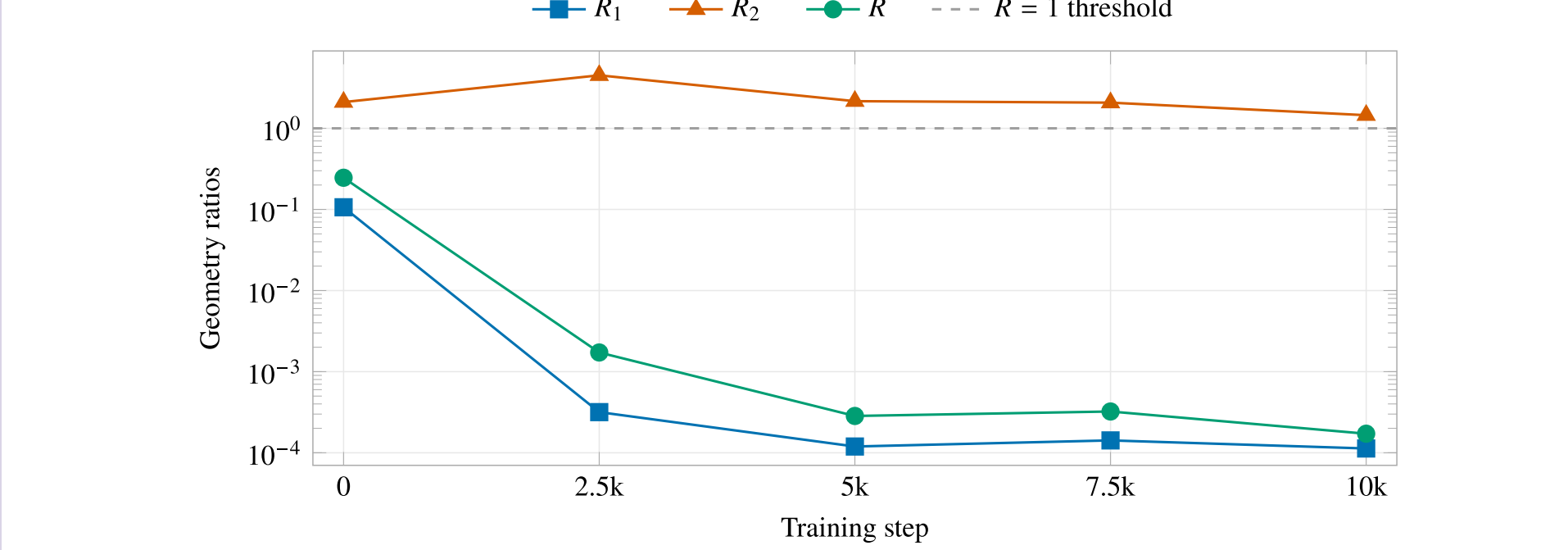
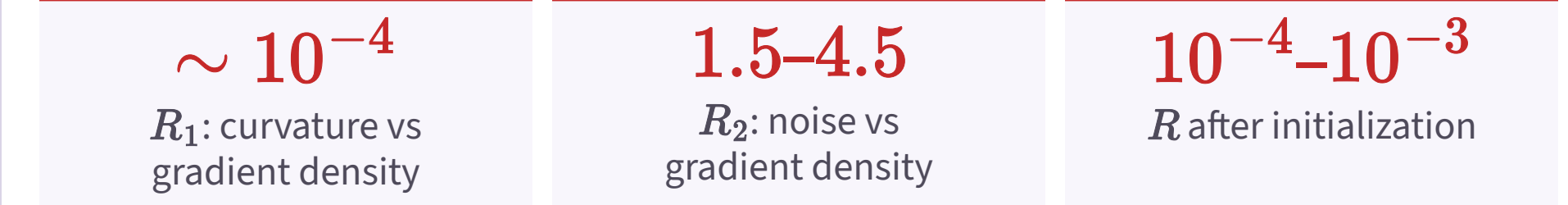


Matrix (A4b): weighted noise moment vs $\|\nabla f\|_*^2$, the fit persists off-trajectory.

A4b linear fit	Natural	+Gaussian	+Student- t
Adam	0.5807	0.9339	0.9465
Lion	0.7986	0.9571	0.9592
Muon	0.8514	0.8678	0.8713

8 Geometry predicted by the theory

In real GPT-2 training, $R \ll 1$ at every checkpoint: the regime where sign descent beats NSGD.

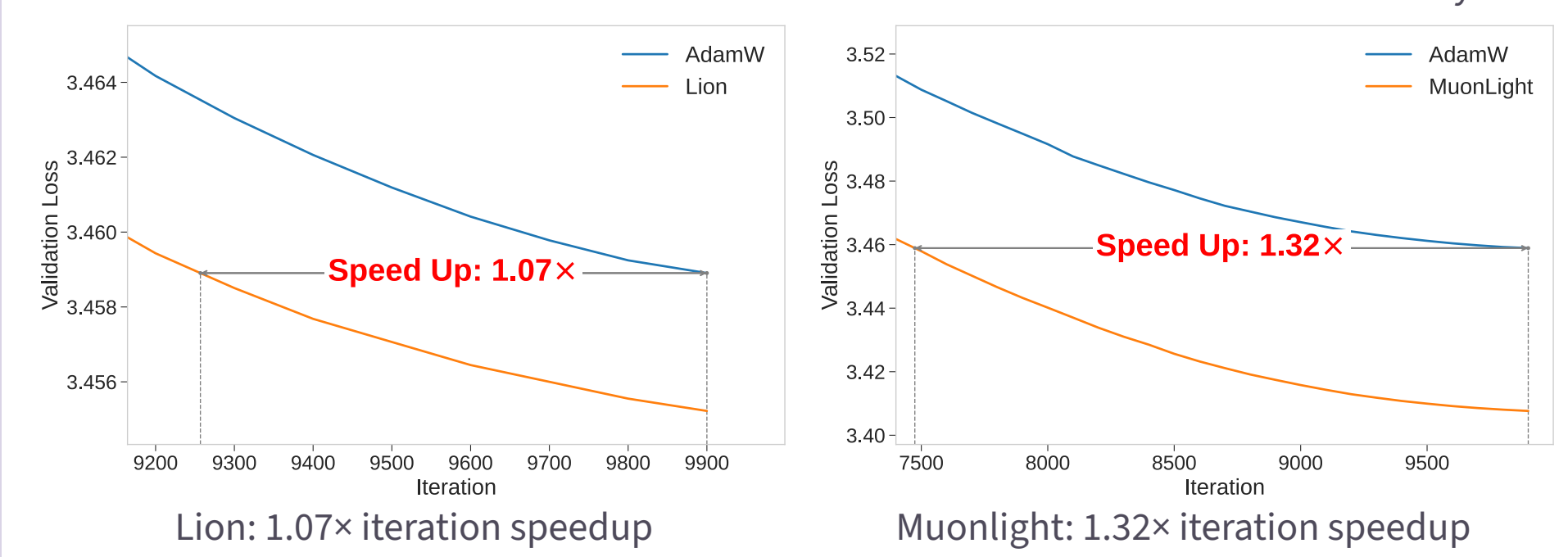
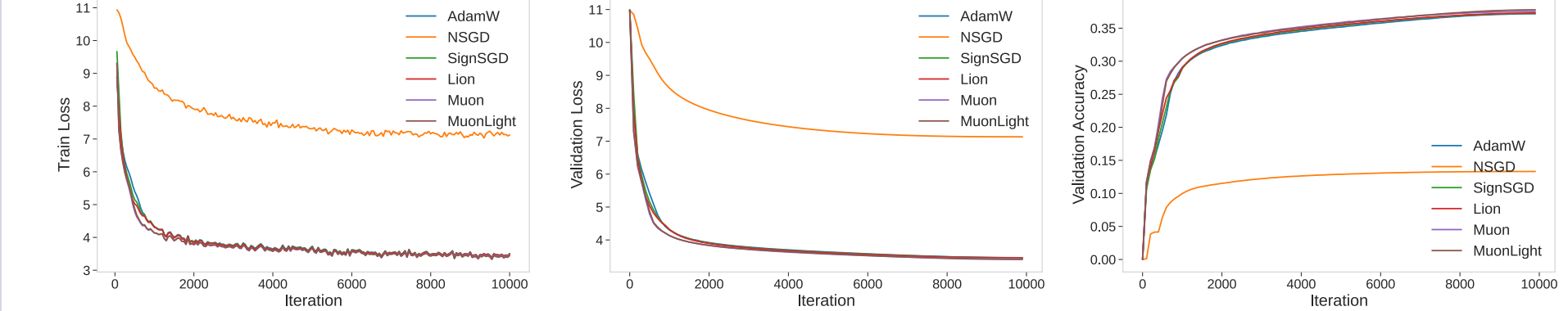


Lion, full model (75 tensors, 124M params); dashed: $R = 1$.

- How to read it:** our bound wins while green R stays under the dashed line. It stays far below: curvature is much more concentrated than the gradient (R_1 tiny), noise only mildly denser ($R_2 > 1$).
- Muon (2304 × 768 attention weight, 10k steps):** noise effective dim. falls $76.4 \rightarrow 37.2$, so $\|\hat{\mathbf{V}}_0\|_* \approx 0.22\sqrt{r} \|\hat{\mathbf{V}}_0\|_F$ ($r = 768$), and its Hessian nuclear / stable rank is only 77.3 / 14.3 (block dim. 1,769,472).

9 Pretraining performance & stability

Same 10k-step budget ($\approx 2.62B$ tokens). Sign-based methods outperform NSGD, and matrix sign methods achieve the best validation performance.



Final val. loss (seeds)	AdamW	Lion	Muon
mean ± std	3.467±0.001	3.464±0.001	3.429±0.0005